



Faculty of Engineering and Computer Science
Department of Electrical and Computer Engineering

USER MANUAL

ELEC/COEN 390 - BB

Electrical/Computer Engineering Product Design Project

Stakeholders

Dr. Wahab Hamou-Lhadj

Team 6

Bilal Samee
27286295

Dominique Reynolds-Sandy
40241168

Navraj Jhajj
40129282

Armaan Khan
40235610

Dipita Sinha
40273009

**We certify that this submission is our original work and meets the
Faculty's Expectations of Originality**

August 15, 2026



Automated Aquarium Monitoring Systems (AAMS Inc.)

AQUASENSE

User Manual • Rev. 1.0.0

Contents

List of Figures	ii
1. Start-up	1
2. Set Up Your Aquarium	2
3. Mounting and Pairing	3
4. Dashboard	4
5. Notifications	6
6. Analytics	7
7. Settings	8

List of Figures

Figure 1 AquaSense Sign-up Screen	1
Figure 2 AquaSense Templates	2
Figure 3 AquaSense Template Advanced Settings	3
Figure 4 AquaSense Mounting Set-up	4
Figure 5 AquaSense Dashboard	5
Figure 6 AquaSense Notification Page	6
Figure 7 AquaSense Analytics Page	7
Figure 8 AquaSense Settings Page	9

1. Start-up

After downloading the app, the sign-up page appears. Please enter your full name and email address, then create a password. Then click Sign up. Alternatively, you can sign in with Google.

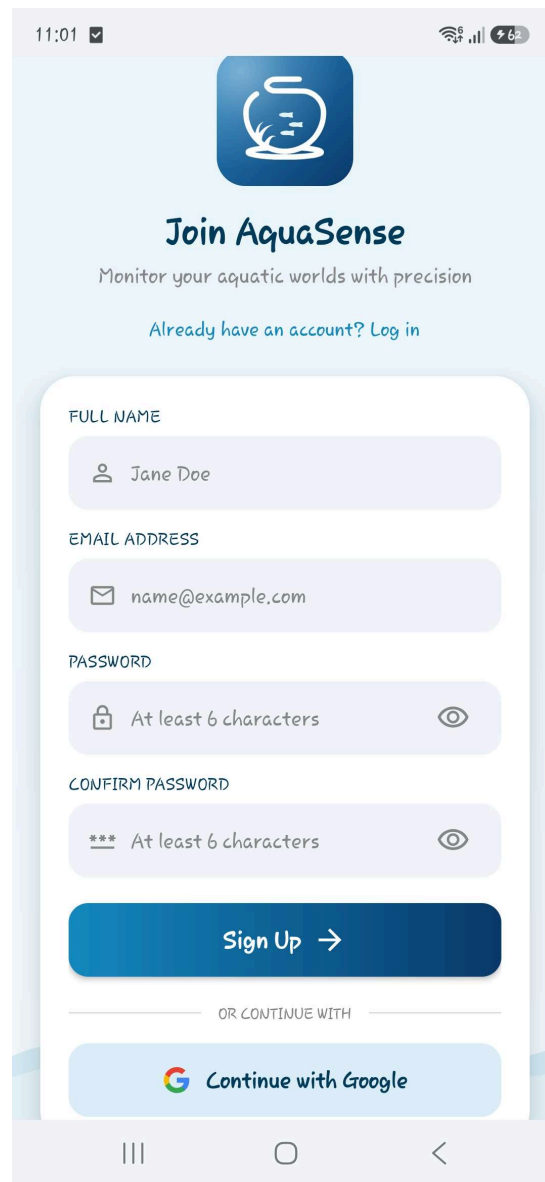
The image shows a mobile app sign-up screen for AquaSense. At the top, there's a status bar with the time 11:01, a checkmark, and signal/battery icons. Below that is the AquaSense logo, a blue square with a white fishbowl icon. The title "Join AquaSense" is in a bold, dark blue font, followed by the tagline "Monitor your aquatic worlds with precision" in a smaller, lighter blue font. A link "Already have an account? Log in" is in a light blue font. The sign-up form has four fields: "FULL NAME" with a person icon and the text "Jane Doe"; "EMAIL ADDRESS" with an envelope icon and the text "name@example.com"; "PASSWORD" with a lock icon, the text "At least 6 characters", and an eye icon; and "CONFIRM PASSWORD" with three asterisks, the text "At least 6 characters", and an eye icon. A large blue "Sign Up →" button is below the fields. Below the button is a horizontal line with the text "OR CONTINUE WITH" in the center. At the bottom is a light blue button with the Google logo and the text "Continue with Google". The bottom of the screen shows the Android navigation bar with three icons: a square, a circle, and a triangle.

Figure 1: AquaSense Sign-up Screen

2. Set Up Your Aquarium

After creating an account, a welcome page appears, giving the next steps to create an aquarium. First, you choose the name for your aquarium. Next, choose a template for the aquarium. Clicking on 'show details' unveils extra information regarding each template. Each aquarium template has different threshold values; decide on a template that best matches your fish's needs.

Optional: Under advanced settings, you can customize the threshold values and spike levels if the given threshold levels aren't accurate enough.

Once the name and template are selected, click Mount Your Hub.

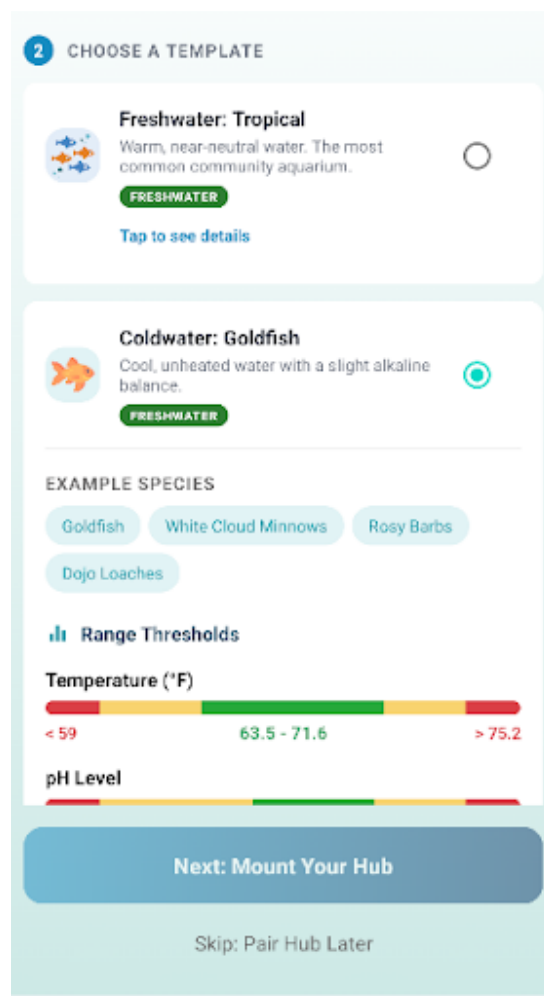


Figure 2: AquaSense Templates

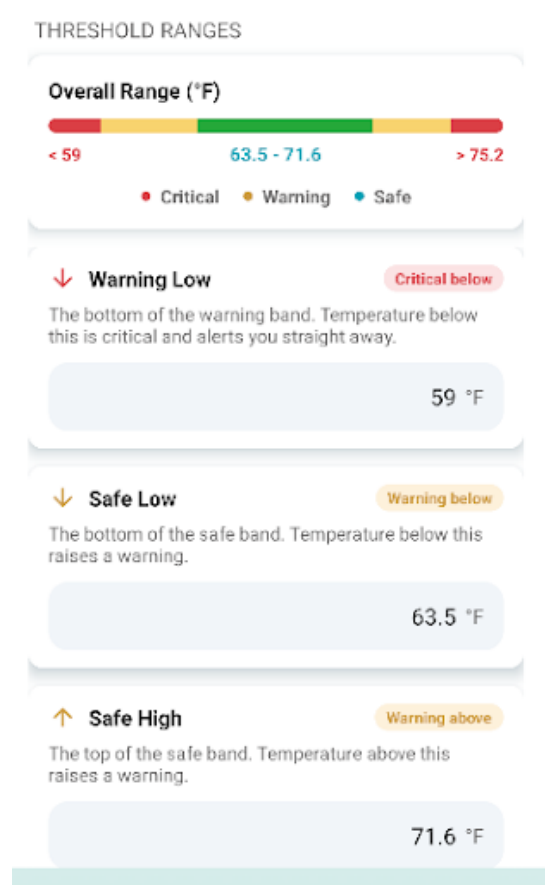


Figure 3: AquaSense Template Advanced Settings

3. Mounting and Pairing

Follow the steps on screen to mount the sensors. Next, pair your hub to the sensors. Make sure Bluetooth and Wifi are turned on.

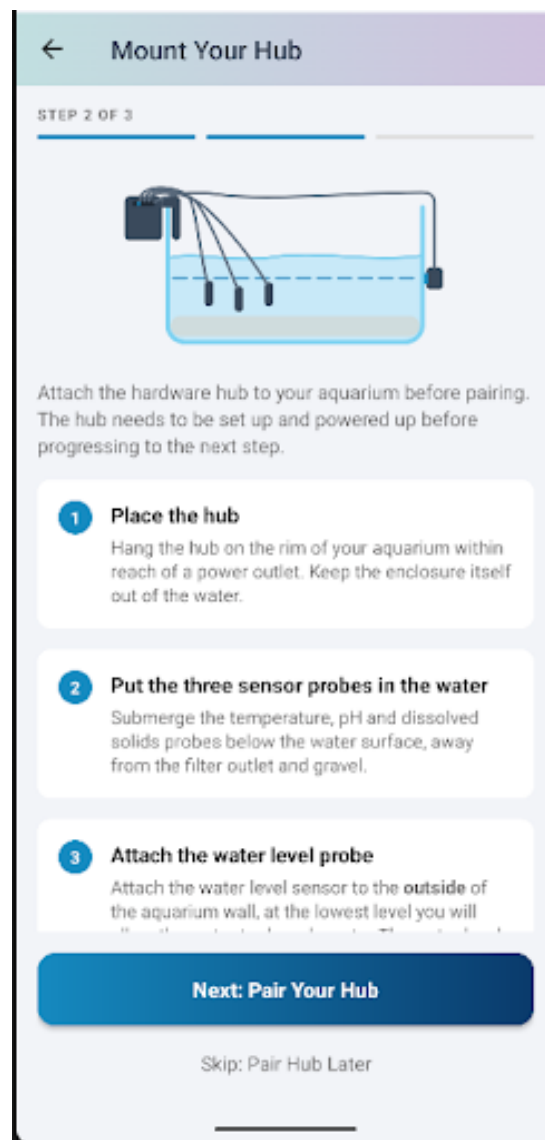


Figure 4: AquaSense Mounting Set-up

4. Dashboard

The main dashboard provides live readings for all connected sensors. In the top-left corner, click 'Aquasense' to switch between your aquariums or create a new profile. The top-right corner displays the active aquarium type.

An emoticon in the center indicates overall system health showing whether parameters are normal, at warning levels, or critical. Below, four dedicated tiles display real-time values for water level, temperature, total dissolved solids (TDS), and pH levels. Clicking the 'i' icon opens

the 'About' section, which includes troubleshooting steps for bringing offline sensors back online.

- Dashboard: Return to the main telemetry view
- Notifications: View real-time alert logs and warning history.
- Analytics: Track historical parameter trends over time.
- Settings: Manage account preferences, network options, and privacy settings.

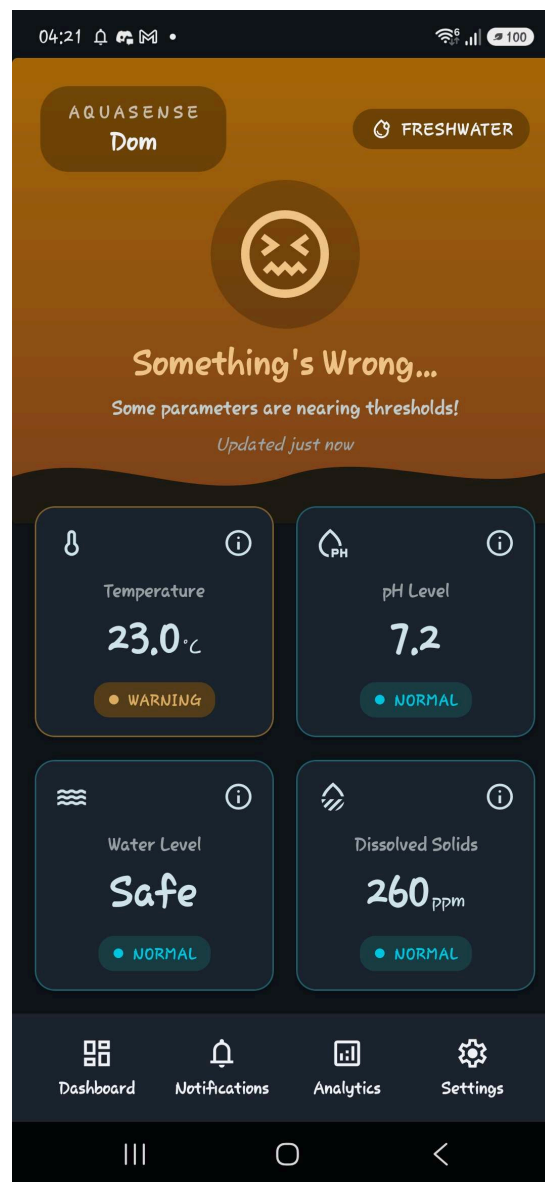


Figure 5: AquaSense Dashboard

5. Notifications

On the notifications tab, you can view all previous alerts alongside the time and date they were sent. Selecting the 'Aquarium' option on the top left allows you to filter notifications by aquarium, while the 'Sensor' option on the top right filters them by a specific sensor. At the bottom right, the 'Clear All' option removes all notifications.

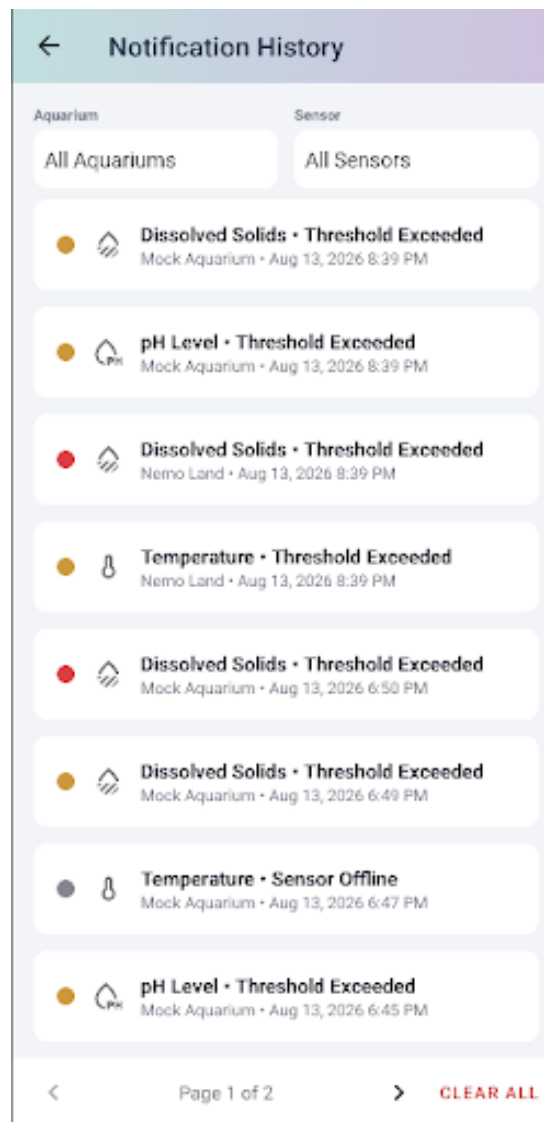


Figure 6: AquaSense Notification Page

6. Analytics

On the analytics page, you can visualize the data from your sensors. In the center of the page, a graph displays all recent parameter trends. You can easily filter the data by selecting specific aquariums, sensors, or time spans. Tapping the blue paper icon above the graph allows you to download a PDF report of your data, while the red trash icon deletes the graph.

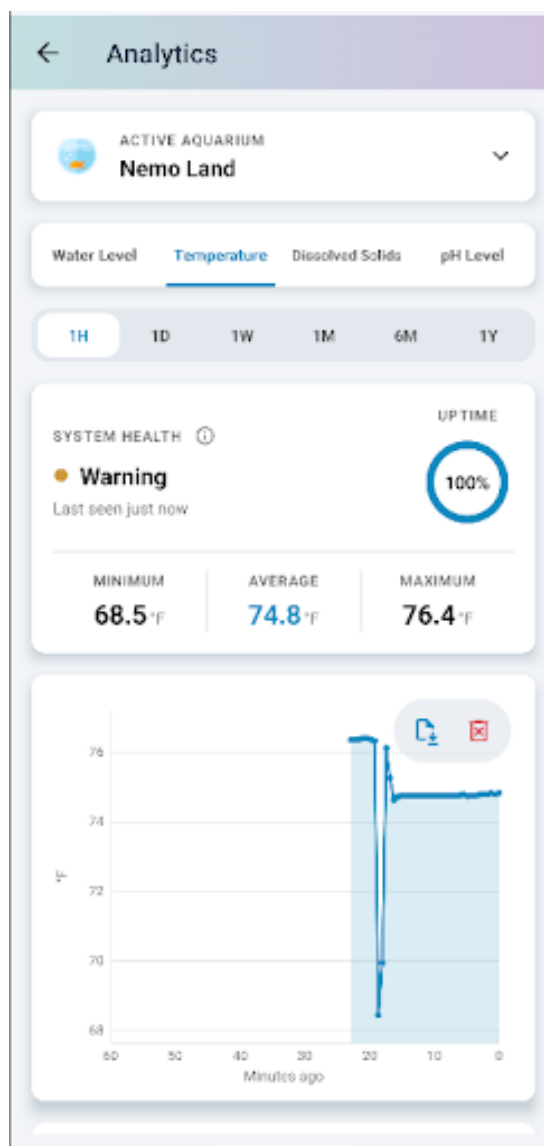


Figure 7: AquaSense Analytics Page

7. Settings

The Settings page is divided into three sections: Aquarium, App & Account, and Help & Support.

- **Aquarium:** Modify your setup by managing aquariums, editing templates, adjusting parameter thresholds, and performing sensor calibrations.
- **App & Account:** Under Display & Units, switch between imperial and metric units or enable dark mode. Under Notifications, configure alert preferences including silencing alerts, selecting update preferences for specific sensors or aquariums, and adjusting spike alerts.
- **Help & Support:** Access the Troubleshooting Guide to view fixes for common issues. Under Contact & Support, visit the help page or find an email address to get in touch. Under Privacy Policy, review your privacy obligations and data guidelines.

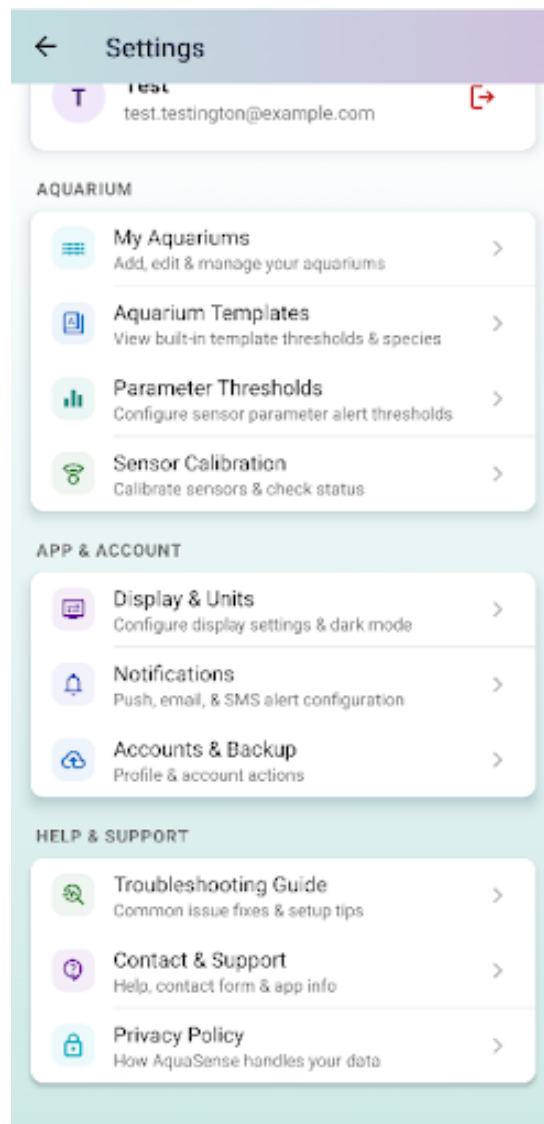


Figure 8: AquaSense Settings Page